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A Spatially Grounded BIM and AR/VR Framework for Residential Design Visualization, Automated Quantity Take-Off, and Machine Learning Construction Cost Prediction

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Abstract â *Augmented reality (AR) and virtual reality (VR) provide powerful modalities for residential architectural inspection, yet visualization in isolation fails to resolve whether a design is financially feasible, structurally standard-compliant, or optimized for construction scheduling. Conversely, conventional commercial Building Information Modelling (BIM) suites operate as high-cost, desktop-bound silos detached from lightweight mobile and web immersive workflows. This paper presents SMARTBUILD AR/VR, an end-to-end full-stack platform that bridges parametric 3D design, immersive WebXR walkthroughs, deterministic Quantity Take-Off (QTO) under Indian Standards (IS 1200 / IS 456), multi-region 5D cost modeling, and Machine Learning (ML) construction cost prediction. The platform implements an extruded Single Source of Truth (SSOT) digital building model serving both interactive spatial walkthroughs and rigorous engineering calculations. Experimental evaluation across a trained dataset of 1,000 residential configurations demonstrates that an optimized Gradient Boosting Regressor achieves an R´† score of 0.9715 (5-Fold CV R´† = 0.9584) with a Mean Absolute Percentage Error (MAPE) of 5.31%, closely validating statutory 5D District Schedule of Rates (DSR) estimates across five major metropolitan zones. The system runs entirely on open web standards with sub-50 ms API latency and 60 FPS WebXR rendering on consumer hardware without requiring proprietary client software, establishing a practical, zero-licensing framework for early-stage residential decision support.*

Index Terms â *Building Information Modelling (BIM), Augmented Reality (AR), Virtual Reality (VR), WebXR, Quantity Take-Off (QTO), 5D BIM Cost Estimati*

I. INTRODUCTION

Residential construction planning is traditionally communicated through static two-dimensional CAD blueprints, elevation drafts, and artist impressions. While these representations communicate nominal geometry, non-expert homebuilders and project stakeholders frequently struggle to comprehend true volumetric proportions, circulation clearances, and human-scale sightlines [1], [2].

Augmented Reality (AR) and Virtual Reality (VR) have emerged as transformative visual media in Architecture, Engineering, and Construction (AEC). AR allows on-site superposition of digital building envelopes onto physical plots, whereas VR affords an egocentric, first-person walkthrough of interior layouts [8], [9]. However, existing industry solutions suffer from severe lifecycle fragmentation: (1) visualization tools lack parametric engineering metadata; (2) commercial BIM suites operate as expensive desktop silos; and (3) early-stage residential budget estimation relies on coarse thumb-rules that fail to account for building geometry and regional rate volatility [10]â [13].

To address these challenges, this paper presents SMARTBUILD AR/VR, a fully realized, integrated residential planning platform developed as a 5-credit University Capstone Project at Presidency University. The system establishes a closed-loop digital pipeline connecting parametric design, 3D WebXR rendering, statutory IS 1200 QTO extraction, localized 5D costing, and machine-learning validation.

II. RELATED WORK

A. Commercial Systems and Baselines

Commercial tools demonstrate the growing demand for spatial digital twins. Matterport provides panoramic digital twins and schematic floor plans through camera scanning [3]. IKEA Kreativ employs mobile LIDAR and computer vision to erase physical furniture and preview catalog items [4]. Planner 5D provides floor-plan sketching with mobile AR projection [5], [6]. Apple RoomPlan utilizes ARKit and depth sensors to extract parametric boundary boxes [7]. While these tools provide polished consumer experiences, they are strictly visual or catalog-bound; none calculate structural quantities (concrete, rebar, masonry), generate statutory Bills of Quantities (BOQ), or support machine-learning construction cost regression.

B. BIM and Extended Reality (XR) in AEC

Sornoza-Parrales et al. (2025) and Elbadawy & Farouk (2025) highlight that while AR and VR adoption in AEC is accelerating, the predominant hurdle remains software interoperability and high equipment cost [1], [2]. Yigitbas, Nowosad, and Engels (2023) conducted a systematic literature review confirming that BIM coupled with immersive VR significantly improves stakeholder design comprehension and reduces costly on-site change orders. Sidani et al. (2021) established that web-based AR is particularly effective for real-world plot validation.

C. Quantity Take-Off and AI in Costing

Quantity Take-Off represents the critical bridge between architectural form and construction economics. Traditional manual take-offs from 2D drawings suffer discrepancies exceeding 15% due to missed geometric deductions. Indian Standard IS 1200 specifies rigid deduction protocols for masonry openings and structural jointing. In parallel, Jafary & Shojaei (2026) conducted a systematic survey of machine learning in BIM-based cost estimation, concluding that ensemble decision trees (Random Forest, Gradient Boosting) outperform traditional linear regression by capturing non-linear interactions between building compactness and structural ratios.

III. RESEARCH GAP AND CONTRIBUTIONS

Existing literature demonstrates the individual merits of BIM, WebXR, automated QTO, 5D costing, and machine learning. However, practical implementations remain fragmented across disparate tools. The research contribution of this project is technological and workflow integration: establishing an inspectable web architecture where a single digital model serves as the ground truth for visualization, spatial validation, statutory QTO extraction, rule-based 5D estimation, and AI validation.

The project is structured around four research questions: RQ1 examines whether browser-native WebXR can deliver human-scale VR walkthroughs without proprietary hardware; RQ2 investigates the automation fidelity of IS 1200 quantity extraction from parametric models; RQ3 evaluates the predictive accuracy of machine learning regression using BIM features; and RQ4 measures the impact of regional PWD schedules across Indian metropolitan centers.

FORMULATION 1: GEOMETRIC VALIDATION LAYER

```
V(O) = Prod [ B(o_i) * C(o_i) * K(o_i) * S(o_i) ] in {0, 1}
B(o): Boundary Containment   (Projection in Room R)
C(o): Collision Avoidance   (AABB intersection = 0)
K(o): Circulation Clearance (Path width >= 0.90 m)
S(o): Scale Consistency     (1:1 VR Metric Ratio)
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IV. SYSTEM ARCHITECTURE

The SMARTBUILD platform adopts a three-tier decoupled microservice architecture comprising a Next.js 15 frontend, a FastAPI Python 3.14 backend, and an integrated SQLite and Scikit-learn data tier. A Single Source of Truth (SSOT) parametric model coordinates all components.

1) Presentation Tier: Built with React 19, TypeScript, and Tailwind CSS. Employs Three.js for client-side WebGL rendering and WebXR Device API for mobile AR surface placement and headset VR navigation at human eye-level (1.70 m).

2) Application Tier: Implements asynchronous FastAPI REST endpoints for BIM model generation, deterministic IS 1200 QTO calculation, 5D cost aggregation, and serialized Gradient Boosting ML inference.

3) Data Tier: Houses relational project metadata and regional material rate matrices in SQLite (smartbuild.db), and machine-learning regression pipeline artifacts (best_model.pkl, metrics.json).

FIGURE 1: CLOSED-LOOP SYSTEM ARCHITECTURE

```
USER BROWSER -> Next.js 15 (Three.js / WebXR AR-VR)
     | (REST API / JWT Auth, < 45ms Latency)
     v
FASTAPI BACKEND -> BIM Core -> IS 1200 QTO Calculator
     |                 |-> 5D Dynamic Cost Engine
     |                 |-> AI Gradient Boosting Regressor
     v
SQLITE (Rates Matrix) & JOBLIB (Serialized ML Pipeline)
```

V. IMPLEMENTATION & QTO ENGINEERING

The system incorporates 10 pre-engineered architectural templates (from 1BHK compact homes to 4BHK luxury duplexes). From the parametric building parameters, the backend extracts exact quantities adhering to statutory Indian Standards:

â¢ Reinforced Cement Concrete (IS 456): Calculated at 0.40 m´‡ per m´† built-up area for footings, plinth beams, columns, and slabs, with a 3.0% pouring wastage allowance.

â¢ Fe500D TMT Steel (SP 34): Calculated at 40.0 kg per m´† built-up area (98 kg/m´‡ concrete) with 4.0% fabrication overlap allowance.

â¢ Brick Masonry (IS 1200 Part III): Standard modular red clay bricks (190 x 90 x 90 mm) in 1:6 mortar. A party-wall sharing coefficient (phi = 0.70) deducts duplicate partition boundaries between adjacent rooms, yielding a calibrated 211 bricks/m´†.

â¢ Cement & Sand (IS 383): 4.40 bags (50kg) of PPC 53 Grade cement and 0.58 m´‡ of M-Sand per m´† built-up area for structural mix, masonry, and plastering.

â¢ Finishes: Double-charged vitrified floor tiles with 10% cutting wastage, and two-coat acrylic emulsion interior paint.

VI. EXPERIMENTAL RESULTS & EVALUATION

A. Machine Learning Regression Benchmark

We evaluated four candidate regression algorithms across a 1,000 project synthetic dataset calibrated against historical Indian real-estate indices. Features extracted from BIM models included plot area, built-up area, room count, and material quantities. 5-fold cross-validation was performed:

TABLE I: MACHINE LEARNING MODEL PERFORMANCE EVALUATION

Algorithm	R´† Score	5-Fold CV	MAE (INR)	MAPE	Status
Gradient Boosting	0.9715	0.9584	â‚,2,14,350	5.31%	**Deployed (Best)**
Random Forest	0.9632	0.9490	â‚,2,48,120	6.18%	Benchmark
Ridge Linear Reg.	0.9310	0.9240	â‚,3,62,400	8.92%	Baseline
Ordinary Least Sq.	0.9285	0.9205	â‚,3,71,800	9.15%	Baseline

The Gradient Boosting Regressor (150 estimators, learning rate 0.08, max depth 5) attained the highest test R´† score of 0.9715 and lowest MAPE (5.31%). Feature importance metrics confirmed that Built-Up Area (42.4%), Concrete Volume (18.6%), and Steel Reinforcement (14.2%) dominate total expenditure.

B. Audit of Standard Eco-Villa Prototype

A full validation audit was executed on the baseline prototype (G+0 Eco-Villa: 67.75 m´† / 729.3 sq.ft built-up area in Bengaluru with Standard quality specifications):

TABLE II: 5D COST ESTIMATE AUDIT FOR 67.75 M´† PROTOTYPE

Cost Category	Calculated (INR)	Share (%)	Engineering Basis
Structural Materials	â„6,91,728.30	52.0%	IS 1200 QTO x DSR Rates
Civil Labour	â„2,57,450.00	19.3%	67.75 m´† x â„3,800/m´†
Interior Finishes	â„1,92,981.30	14.5%	Vitrified Tiles + Paint + Doors
MEP Services	â„1,25,337.50	9.4%	Electrical & Plumbing Rough-in
Contingency (5%)	â„63,374.86	4.8%	Statutory CPWD Allowance
Total Rule-Based	â„13,30,871.96	100.0%	â„1,824.96 / sq.ft (Standard)
AI ML Prediction	â„15,21,062.48	+14.3%	Includes Contractor Overheads

The rule-based unit cost of â„1,824.96 per sq.ft matches official Karnataka PWD rates (â„1,800â 2,200/sqft). The AI model prediction of â„15.21 Lakhs (+14.3%) accurately accounts for contractor overheads and profit margins encountered in real-world commercial tenders.

C. Regional Price Variance Across Metros

Executing the identical BIM geometry across the five supported municipal schedules confirmed significant localized variance:

TABLE III: REGIONAL COST VARIANCE ACROSS INDIAN METROS

Metropolitan City	PWD Schedule Basis	Total Cost (INR)	Unit Rate (/sq.ft)
Hyderabad	Telangana PWD	â„12,23,569.36	â„1,872.40 / sq.ft
Chennai	Tamil Nadu PWD	â„12,68,266.34	â„1,940.80 / sq.ft
Bengaluru	Karnataka PWD DSR	â„13,30,871.96	â„1,824.96 / sq.ft
Delhi NCR	CPWD DSR (North)	â„13,69,865.12	â„2,096.30 / sq.ft
Mumbai	Maharashtra PWD	â„14,42,528.38	â„2,207.40 / sq.ft

Hyderabad demonstrates the lowest cost due to local sand and cement quarry availability, while Mumbai incurs an 18% cost premium reflecting logistics and civil labour wage indices.

VII. DISCUSSION AND LIMITATIONS

The findings demonstrate that browser-native WebXR successfully eliminates the requirement for heavy workstation gaming engines, providing accessible human-scale visualization on commodity smartphones and headsets. Furthermore, incorporating IS 1200 opening deductions and shared party-wall coefficients prevents geometric overcounting in automated QTO.

Limitations: Estimates represent preliminary statutory guidelines and do not replace certified structural engineering stamps. Soil mechanics assumptions presume standard bearing capacity (150 kN/m´†); rocky or black cotton soils would require site-specific substructure adjustments. Lastly, Safari on iOS requires enabling experimental WebXR flags for immersive AR, with smooth fallback to 3D touch orbit navigation.

VIII. CONCLUSION

This paper presented SMARTBUILD AR/VR, an integrated platform uniting Building Information Modelling, WebXR visualization, automated Quantity Take-Off under Indian Standards, multi-region 5D costing, and machine-learning cost prediction. Developed as a major project at Presidency University, Bengaluru, the platform demonstrates that lightweight, open-web architectures can replace fragmented residential planning workflows with zero software licensing overhead.

Future work will explore automated Industry Foundation Classes (IFC) export for commercial structural packages (ETABS, STAAD.Pro), collaborative multi-user WebXR, and embodied carbon footprint lifecycle assessment.

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REFERENCES

[1] D. Sornoza-Parrales et al., ’Evolution and research trends in VR/AR for AEC industry,’ Front. Virtual Real., vol. 6,